

679 final project code

```

Rows: 1,000
Columns: 9
$ Order_ID          <int> 522, 738, 741, 661, 412, 679, 627, 514, 860, 13~
$ Distance_km       <dbl> 7.93, 16.42, 9.52, 7.44, 19.03, 19.40, 9.52, 17~
$ Weather           <chr> "Windy", "Clear", "Foggy", "Rainy", "Clear", "C~
$ Traffic_Level     <chr> "Low", "Medium", "Low", "Medium", "Low", "Low",~
$ Time_of_Day       <chr> "Afternoon", "Evening", "Night", "Afternoon", "~
$ Vehicle_Type      <chr> "Scooter", "Bike", "Scooter", "Scooter", "Bike"~
$ Preparation_Time_min <int> 12, 20, 28, 5, 16, 8, 12, 5, 20, 29, 13, 29, 12~
$ Courier_Experience_yrs <dbl> 1, 2, 1, 1, 5, 9, 1, 6, 6, 1, 4, 1, 5, 7, NA, 3~
$ Delivery_Time_min  <int> 43, 84, 59, 37, 68, 57, 49, 46, 35, 73, 88, 76,~

```

Order_ID	Distance_km	Weather	Traffic_Level
Min. : 1.0	Min. : 0.590	Length:1000	Length:1000
1st Qu.: 250.8	1st Qu.: 5.105	Class :character	Class :character
Median : 500.5	Median :10.190	Mode :character	Mode :character
Mean : 500.5	Mean :10.060		
3rd Qu.: 750.2	3rd Qu.:15.018		
Max. :1000.0	Max. :19.990		

Time_of_Day	Vehicle_Type	Preparation_Time_min
Length:1000	Length:1000	Min. : 5.00
Class :character	Class :character	1st Qu.:11.00
Mode :character	Mode :character	Median :17.00
		Mean :16.98
		3rd Qu.:23.00
		Max. :29.00

Courier_Experience_yrs	Delivery_Time_min
Min. :0.000	Min. : 8.00
1st Qu.:2.000	1st Qu.: 41.00
Median :5.000	Median : 55.50

Mean	:4.579	Mean	: 56.73
3rd Qu.:	7.000	3rd Qu.:	71.00
Max.	:9.000	Max.	:153.00
NA's	:30		

Table 1: Missing value counts in raw dataset.

Variable	Missing
Courier_Experience_yrs	30

[1] 30

[1] 970

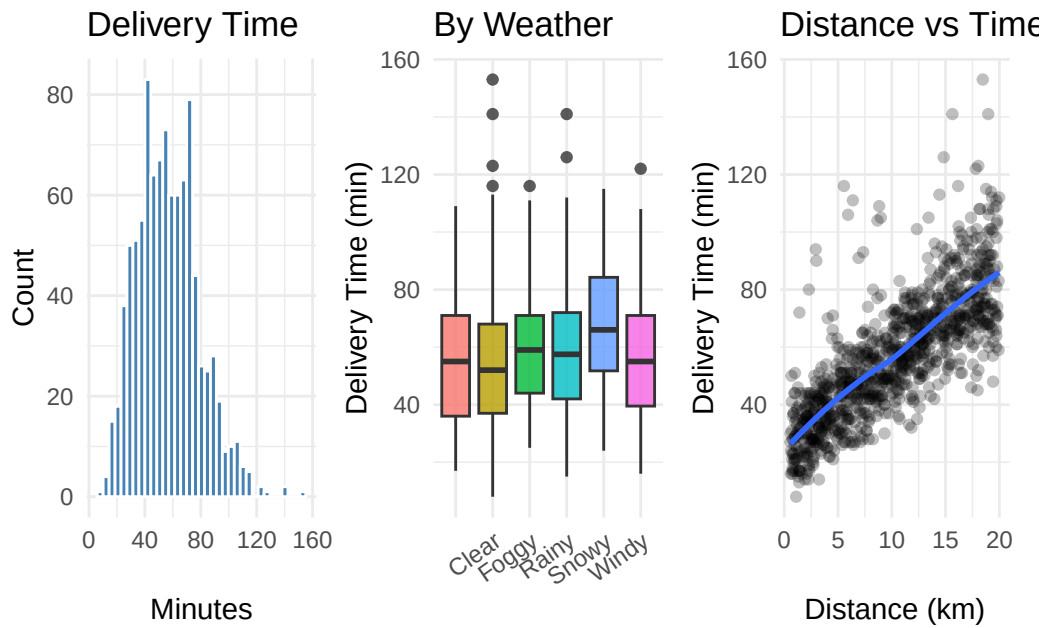
[1] 970

	Order_ID	Distance_km	Weather	Traffic_Level	Time_of_Day	Vehicle_Type
1	814	18.46	Clear			Scooter
2	446	18.97	Clear	Low	Evening	Car
3	394	15.64	Rainy	Low		Bike
4	385	14.83	Rainy	Low	Morning	Car
5	948	18.05	Clear	Medium	Evening	Scooter
6	428	17.81	Windy	High	Evening	Bike

	Preparation_Time_min	Courier_Experience_yrs	Delivery_Time_min
1	29		153
2	25		141
3	20		141
4	19		126
5	10		123
6	21		122

[1] 778

[1] 192



Call:

```
lm(formula = Delivery_Time_min ~ Distance_km + Preparation_Time_min +
    Courier_Experience_yrs + Weather + Traffic_Level + Time_of_Day +
    Vehicle_Type, data = train_data)
```

Residuals:

Min	1Q	Median	3Q	Max
-22.932	-5.096	-1.140	2.810	61.794

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	26.90861	3.90805	6.885	1.21e-11	***
Distance_km	2.97303	0.06668	44.585	< 2e-16	***
Preparation_Time_min	1.00388	0.05386	18.638	< 2e-16	***
Courier_Experience_yrs	-0.62652	0.13031	-4.808	1.84e-06	***
WeatherClear	-6.17174	2.24814	-2.745	0.006189	**
WeatherFoggy	2.06233	2.48883	0.829	0.407571	
WeatherRainy	-0.56217	2.33605	-0.241	0.809891	
WeatherSnowy	3.49617	2.51406	1.391	0.164740	
WeatherWindy	-4.24310	2.50805	-1.692	0.091096	.
Traffic_LevelHigh	2.86108	2.30307	1.242	0.214514	
Traffic_LevelLow	-8.21047	2.21469	-3.707	0.000225	***
Traffic_LevelMedium	-2.58020	2.21536	-1.165	0.244512	

```

Time_of_DayAfternoon    -7.97355    2.20344   -3.619  0.000316 ***
Time_of_DayEvening      -7.39179    2.19991   -3.360  0.000818 ***
Time_of_DayMorning      -8.28838    2.20921   -3.752  0.000189 ***
Time_of_DayNight        -9.11375    2.47381   -3.684  0.000246 ***
Vehicle_TypeCar          -0.14927    1.02010   -0.146  0.883702
Vehicle_TypeScooter     -0.43201    0.88000   -0.491  0.623625

```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 10.56 on 760 degrees of freedom

Multiple R-squared: 0.7801, Adjusted R-squared: 0.7752

F-statistic: 158.6 on 17 and 760 DF, p-value: < 2.2e-16

Call:

```
randomForest(formula = Delivery_Time_min ~ Distance_km + Preparation_Time_min +
```

Courier

```
  Type of random forest: regression
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    Number of trees: 500
```

```
No. of variables tried at each split: 3
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```

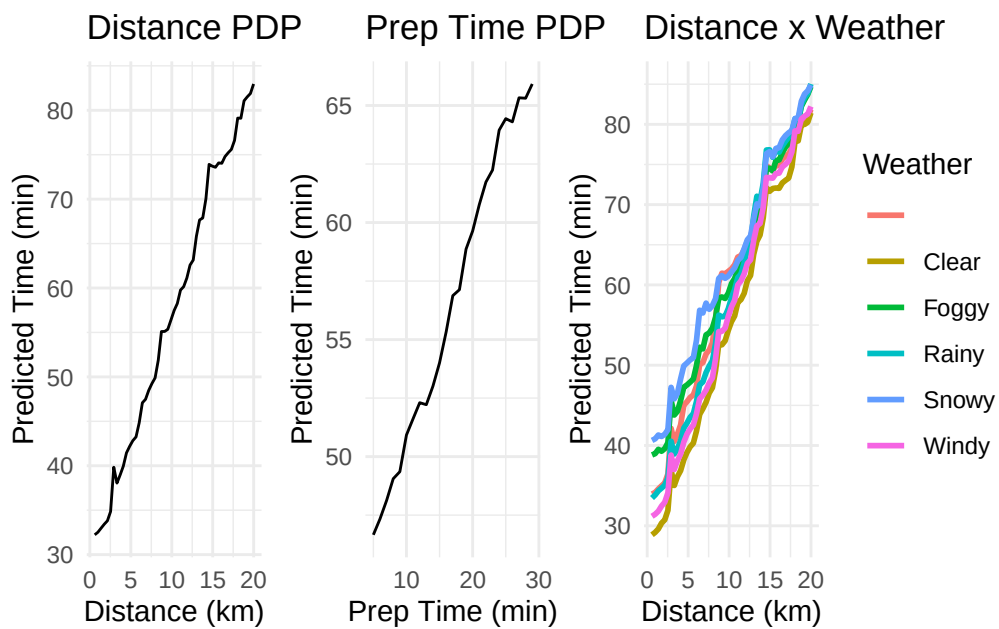
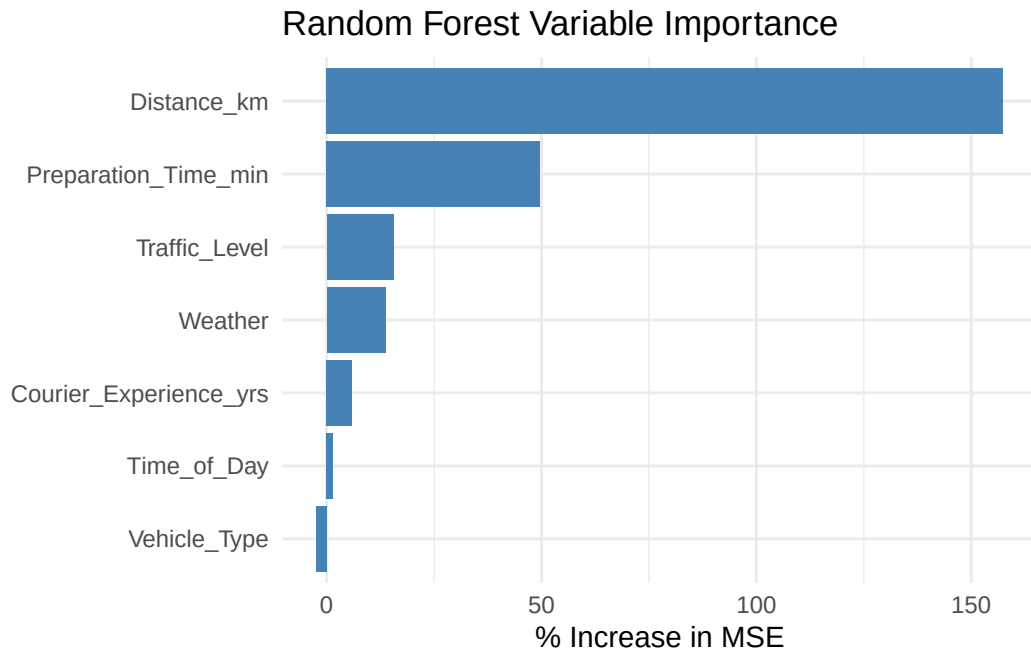
      Mean of squared residuals: 135.7945
        % Var explained: 72.58

```

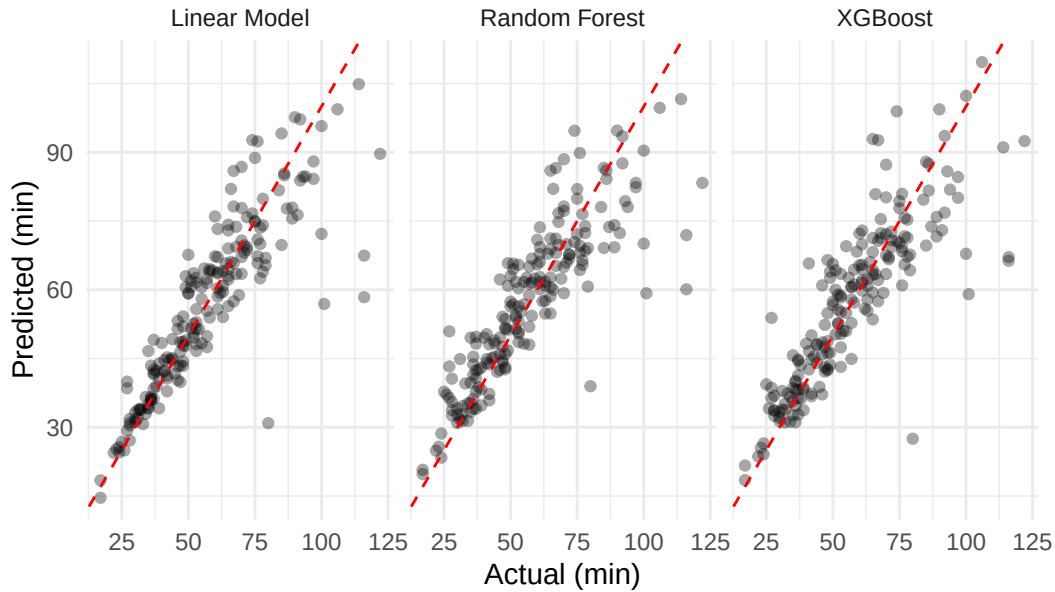
NULL

Table 2: Out-of-Sample Model Performance

Model	RMSE	MAE	R2
Linear Model	10.424	6.459	0.760
Random Forest	11.055	7.564	0.730
XGBoost	11.348	7.503	0.715

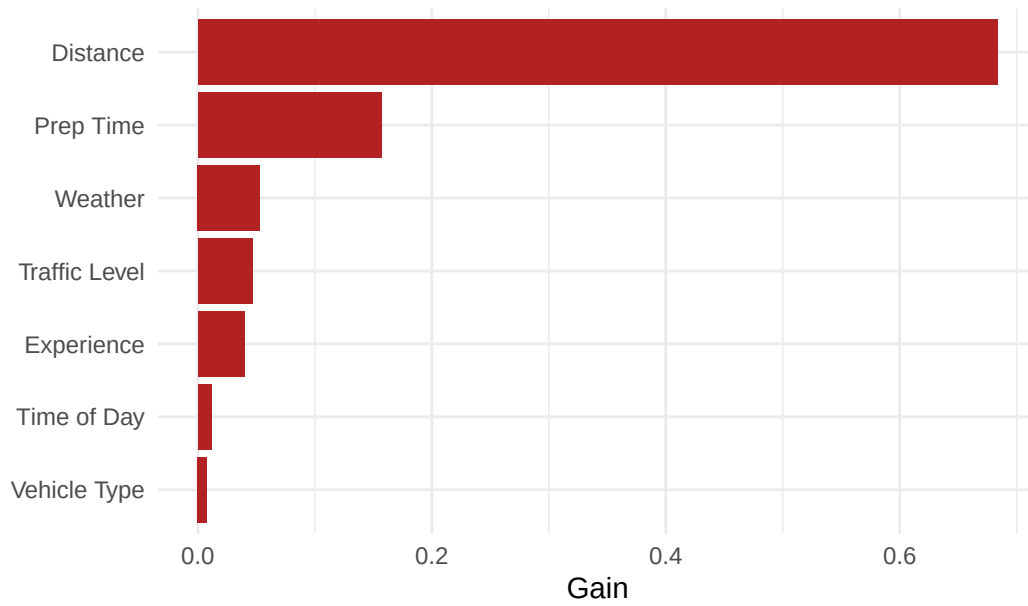


Predicted vs Actual Delivery Time



	Feature <char>	Gain <num>	Cover <num>	Frequency <num>
1:	Distance_km	0.683676860	0.412606683	0.31080438
2:	Preparation_Time_min	0.157262063	0.209630077	0.22941456
3:	Courier_Experience_yrs	0.039928705	0.106241992	0.13993337
4:	Traffic_LevelHigh	0.025660442	0.039886928	0.03950500
5:	WeatherClear	0.018943893	0.035714528	0.03807711
6:	Traffic_LevelLow	0.018776110	0.043574639	0.03950500
7:	WeatherFoggy	0.012255689	0.020567121	0.02950976
8:	WeatherRainy	0.011223114	0.022014412	0.02332223
9:	WeatherSnowy	0.009046021	0.022221168	0.02284626
10:	Vehicle_TypeCar	0.004428473	0.012188426	0.02094241
11:	Time_of_DayEvening	0.003659336	0.007280517	0.01808663
12:	Time_of_DayAfternoon	0.003636621	0.011446138	0.01761066
13:	Vehicle_TypeScooter	0.003418028	0.008338022	0.01713470
14:	Time_of_DayMorning	0.002964137	0.009500600	0.01237506
15:	Traffic_LevelMedium	0.001978652	0.006938183	0.01094717
16:	WeatherWindy	0.001685509	0.018279249	0.01475488
17:	Time_of_DayNight	0.001456349	0.013571317	0.01523084

XGBoost Variable Importance



Random Forest OOB RMSE

